Categorical approach to Cross-documents: Extension to Democritus

CMPSCI 692CT: Category Theory for AGI

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Current System

What Democritus Does:

- Retrieves scientific papers from web
- Extracts stated causal claims into database
- Group claims by the amount of papers supporting it
- Answers which claims appear across documents

What new project extension adds:

- If claim A and B causes claim C, does A cause C?
- When two papers disagree, what does that reveal?
- What category theory concepts can we use to force such causal structures and relationships?
- Possibility of finding the causal truth that no paper explicitly stated

Extension approach: Two Categorical Modules

Right Kan Extension for chain inference:

- If doc 1 says A → B and doc 2 says B→C, then we infer A→C with conservative confidence
- even if no paper ever wrote a sentence of A→C
- geometric-mean Confidence: √(p x q)

Dually, the ***right Kan extension*** $\mathrm{Ran}_K F$ *is the universal extension characterized by maps*

$$\epsilon : (\mathrm{Ran}_K F) \circ K \Rightarrow F.$$

$$\begin{tikzcd} \mathcal{C} \arrow[rr, "F"] \arrow[d, "K"'] & & \mathcal{E} \\ \mathcal{D} \arrow[urr, bend left, "\mathrm{Lan}_K F"] \arrow[urr, bend right, "G"'] \arrow[urr, phantom, "\Rrightarrow \eta"] & & \end{tikzcd}$$

Sheaf colimit for resolving conflicting claims:

- When doc 1 says A→B and doc 2 says B→A, rather than ignoring the information, we resolve conflict through bidirectional feedback loop A ↔B
- concept applied to harmonic-mean confidence

Here, the harmonic-mean confidence of the feedback loop is:

$$conf(A \leftrightarrow B) = \frac{2pq}{p+q}, p = conf(A \rightarrow B), q = conf(B \rightarrow A)$$

Category Theory understanding

Functor

- Structure preserving map between categories
- Here, maps our corpus of documents → confidence scores for causal claims

Kan Extension

- extends partial function to new inputs using universal properties
- Fills in novel A→ C edges we haven't seen using existing A→ B and B→ C

$$A \xrightarrow{Relation_1} B \text{ (from document } S_1\text{, confidence p)}$$

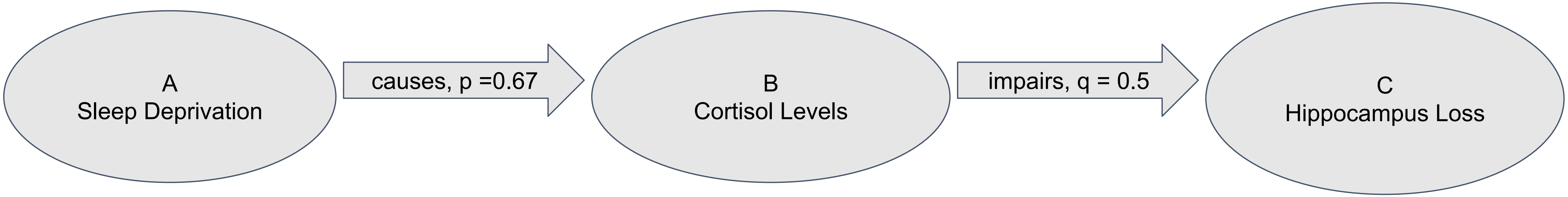
$$B \xrightarrow{Relation_2} C \text{ (from document } S_2\text{, confidence q)}$$

$$A \xrightarrow{Relation_1 + Relation_2} C \text{ (geometric-mean confidence } = \sqrt{p \cdot q}\text{)}$$

Sheaf Colimit

- glues together local, possibly conflicting pieces of data
- where A → B vs B → A across papers can become A ↔ B feedback loop depending on confidence

Kan Extension chaining example



Inferred Kan extension:

Sleep Deprivation → Hippocampus Loss

Given source papers, it may not have written this claim before. Confidence = √(0.67 x 0.5) = 0.58

$$(\mathrm{Ran}_J F)(A \to C) = \sqrt{\mathrm{conf}(A \to B) \times \mathrm{conf}(B \to C)} = \sqrt{p \cdot q}$$

Sheaf Colimit conflict resolving

PAPER 1 states:
Sleep Deprivation $\longrightarrow$ Cognitive Performance
(p=0.67)

CONFLICT

PAPER 2 states:
Cognitive Performance $\longrightarrow$ Sleep Deprivation
(p=0.33)

Sheaf colimit resolves to:

$$conf(A \leftrightarrow B) = \frac{2pq}{p+q}, p = conf(A \rightarrow B), q = conf(B \rightarrow A)$$

Sleep deprivation $\leftrightarrow$ Cognitive Performance (bidirectional feedback loop)

Confidence = 2pq/(p+q) = 0.44, neither papers stated this bidirectionally

Architecture

1. User query to CLIFF: user submits query in browser. CLIFF then routes it to democritus worker
2. Democritus retrieves paper: downloads relevant papers, extracts causal claims as edges in CSQL
3. Domain filtering: keeps only edges whose concepts overlap with query to avoid off-topic drifts
4. Kan Extension module: find A→B→C chain with geometric mean confidence
5. Sheaff colimit model: detect A→B vs B→A conflicts, resolve feedback loop depending on harmonic mean confidence
6. Output dashboard: Json summary and HTML dashboard opened in browser

Two of the newly implemented files were added in the `functorflow_v3` package:

- `democritus_discovery.py` - main categorical modules (Kan extension and Sheaf colimits), SQLite reader and JSON/HTML output
- `run_causal_discovery.py` - high level processor that locates CSQL database and readers run metadata, starts democritus discovery, and opens browser

```
[runner] Running categorical discovery …
[discovery] Auto-detected domain filter: 33 tokens.
[discovery] Raw edges: 401  →  Domain-filtered: 177
[discovery] MIN_OVERLAP=2: 83/177 edges have ≥1 successor
  BRIDGE  apnea → risk of hypertension and cardiac arrhyth
    SUCC  obesity increases the risk of developing hypertens
  BRIDGE  autoimmune activity in rheumatic conditions → risk of cardiovascular complications
    SUCC  cushing s syndrome increases the risk of chronic m
  BRIDGE  autoimmunity increases inflammation which → progression of joint damage in rheumatic
    SUCC  progression of chronic diseases
[discovery] Sheaf colimit: 15576 pairs checked, 15531 failed overlap≥1, 31 conflicts found.

  Documents       : 3
  Stated edges    : 177
  Kan Extension   : 40
  Sheaf Colimit   : 20
```

Experiment and Results

Query: Find me 10 studies in the bidirectional relationship between sleep deprivation and cortisol, including papers that disagree on the primary causal direction.

- Source documents: 3
- Edges after domain filtering: 401→177
- Kan extension (novel claims): 40
- Sheaf colimit (feedback loop): 20
- Total possible novel claims found: 60

Additional experiments:

GLP-1 Receptor Agonists Correctly linked GLP-1 drugs → obesity → insulin resistance → cardiovascular risk
Depression → Cardiovascular Identified cortisol as bridging mechanism between chronic stress and cardiac risk
Sleep & Thyroid Function Confirmed bidirectional relationships consistent with primary sleep corpus query

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Limitations and future work

- Small corpus - only 3 documents retrieved for the sleep query experiment, more documents would produce a better conflict structure and higher confidence, possibly limited due to token query limits or guardrails
- Topic Drift - democritus can retrieve off-topic papers, domain filtering was added to this project, however, still limited and needs stricter corpus containment if we need more
- Chain depth = 1 - Kan extension only chains A→B→C (depth 1), but could have deeper chains A→B→C→D which can reveal more, but will put at risk of hallucination
- Confidence Calibration - geometric mean is conservative but approximate, not as accurate. Therefore, needing longer chain (as said above) with shared evidence for better weighing factors

Demo

THANK YOU!

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